Level 3: Advanced Topics

Specialised methods for complex clinical data: workshops and consultancy

Organisation: Neudata · *#ClearDataClearImpact*

> **Status: outline. Modules are developed on request, after Level 1 and Level 2 have run.**

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1. What Level 3 is

Real clinical data are often more complex than the cohort used in Levels 1 and 2: repeated measurements on the same patients, patients clustered in wards or hospitals, competing outcomes, large routine registries, or questions about prediction and diagnosis rather than association. Level 3 offers specialised modules, each tied to the kind of data or question participants meet in their own work. It also covers the research-craft topics that were moved out of Level 2 so that Level 2 could focus on statistical models: reproducible projects, version control, Quarto reports, figures that meet journal requirements, and missing-data and sensitivity analyses in depth (Modules G and H). Neudata runs them as workshops and consultancy: a short, intensive teaching block followed by hands-on work on the participant's own project, with Neudata support until a defensible analysis plan or analysis is in place.

2. Format

Who	Level 2 graduates (or researchers with equivalent R and regression experience), usually 3 to 10 per workshop; individual researchers or project teams for consultancy.
Prerequisites	**Level 2** (or equivalent): working with R scripts, data cleaning, Table 1, and linear, logistic, count (Poisson, negative binomial, zero-inflated) and Cox models with their assumptions and diagnostics; the basics of missing data. A real question or dataset is strongly recommended.
Workshop format	1 or 2 online sessions of 3 hours: concepts, then a worked example in R, then participants apply it to their own data. The class works together as one group (no breakout rooms); the applied work is individual, and participants share results back to the class.
Consultancy format	A scoping meeting, then 2 to 4 online follow-up meetings over 1–3 months; the participant does the analysis, Neudata reviews the plan, code and interpretation.
Software	R and RStudio; module-specific packages (named per module). Every module uses ggplot2 for its EDA and result figures.
Output	A statistical analysis plan, a reviewed analysis script and draft Methods/Results text for the participant's project, or a structured appraisal for the appraisal-focused modules.

How to request	The department or a project team contacts Neudata with a short description of the question and data; Neudata proposes the module(s) and the format.

3. Principles for every module

These apply to every Level 3 module and every session within it.

1. **An EDA section in every lecture.** Each session has a section on exploratory data analysis for its analysis problem: which summaries and plots to look at before fitting the model, and what they can reveal. Examples: spaghetti plots and missingness over time for repeated measures (Module E); follow-up and censoring patterns for survival (Module D); the distribution of the index test by disease status (Module C); study characteristics and effect sizes before pooling (Module A).
2. **Figure code in every lecture.** Each session gives ready-to-use ggplot2 code for the figures that its problem needs, built to journal standard from the start. Examples: forest and funnel plots (A), calibration plots (B), ROC curves (C), cumulative incidence curves (D), individual trajectories and model-based predictions (E). The general rules for journal figures (vector vs raster files, dimensions, resolution and fonts) are taught once in Module G and then applied in every module.
3. **Reproducible from the start.** Module work is done in a reproducible project (Module G), so that someone else can re-run the analysis and the report.

4. The modules

Module A: Systematic review and meta-analysis

Aims. Participants will be able to (1) write and register (PROSPERO) a systematic-review protocol; (2) build a reproducible multi-database search and screen records in pairs; (3) extract data and assess risk of bias (RoB 2, ROBINS-I); (4) pool effect estimates with fixed- and random-effects models, interpret heterogeneity (I^2, prediction intervals) and draw forest and funnel plots; and (5) report to PRISMA 2020 and grade certainty (GRADE).

Prerequisites. Level 2; a review question (PICO) agreed with Neudata. Packages: meta, metafor.

Session	Content	Product
A1: Protocol, search and screening	Review question and eligibility; PROSPERO; search strategy across databases; dual screening; PRISMA flow diagram; data-extraction form	Registered (or ready-to-register) protocol and search strategy
A2: Bias, pooling and reporting	Risk-of-bias assessment; pooling ORs/RRs/mean differences in R; fixed vs random effects; I^2, subgroups and meta-regression (introduced); small-study effects and funnel plots; GRADE; PRISMA 2020	Forest plot and draft Results for the participant's review

Module B: Prediction models and machine learning

Aims. Participants will be able to (1) distinguish prediction from causal questions; (2) calculate the sample size needed to develop a model; (3) develop a logistic or survival prediction model, assess

discrimination and calibration, and validate it internally (bootstrapping); (4) understand what machine-learning methods add and where they mislead (data leakage, overfitting, fairness); and (5) report with TRIPOD+AI and appraise with PROBAST.

Prerequisites. Level 2 (Session 3, logistic regression in depth). Packages: rms, pmsampsize, pROC, glmnet, tidymodels (introduced).

Session	Content	Product
B1: Building a model properly	Prediction vs causation; candidate predictors; sample size for prediction models; development on FLUID-ICU or own data; discrimination and calibration plots; bootstrap internal validation; presenting the model (nomogram, risk chart)	Model report with optimism-corrected AUC and a calibration plot
B2: Machine learning and appraisal	Penalised regression and a tree-based method compared with logistic regression; data leakage; external validation and updating; fairness; appraising a published AI/ML model with TRIPOD+AI and PROBAST	Structured appraisal of a published model, or a validation plan for one's own model

Module C: Diagnostic-accuracy studies

Aims. Participants will be able to (1) design a diagnostic-accuracy study (index test, reference standard, spectrum, blinding, verification); (2) calculate sensitivity, specificity, predictive values and likelihood ratios with 95% CIs, and draw ROC curves with AUC; (3) choose and justify a cut-off; (4) compare two tests on the same patients; and (5) report to STARD 2015 and appraise with QUADAS-2.

Prerequisites. Level 2. Packages: epiR, pROC.

Session	Content	Product
C1: Design, analysis and reporting	Diagnostic questions and designs; spectrum, verification and incorporation bias; accuracy table with CIs; predictive values across prevalences; ROC/AUC and cut-offs (e.g. lactate for 28-day mortality); comparing two tests (paired ROC); STARD 2015 and QUADAS-2	Completed accuracy analysis and a STARD-based study outline
C2 (optional): Your own diagnostic question	Protocol for a unit-level accuracy study; sample size for sensitivity and specificity; data-collection form	Protocol outline

Module D: Advanced survival analysis

Aims. Participants will be able to (1) go beyond the Level 2 Cox model when its assumptions fail (non-proportional hazards, time-varying effects); (2) use time-varying covariates and avoid immortal-time bias; (3) analyse competing risks (cumulative incidence functions, cause-specific and Fine–Gray models), for example ICU discharge alive vs death; (4) present restricted mean survival time as an alternative to the HR; and (5) report time-to-event analyses clearly.

Prerequisites. Level 2, including Session 6 (Kaplan-Meier, the Cox model and the proportional-hazards check). Packages: survival, survminer, tidycmprsk, cmprsk.

Session	Content	Product
D1: Beyond the basic Cox model	EDA for time-to-event data with changing exposures; when proportional hazards fail: stratification and time-varying effects; time-varying covariates and the counting-process format; immortal-time bias and landmark analysis; restricted mean survival time	Cox analysis with PH checks and a time-varying covariate
D2: Competing risks	Why 1 − KM overestimates risk with competing events; cumulative incidence; cause-specific vs Fine–Gray models and when to use each; plotting and reporting	Competing-risk analysis of FLUID-ICU or own data

Module E: Longitudinal data and multilevel (mixed) models

Aims. Participants will be able to (1) structure repeated-measures data (long vs wide) and plot individual trajectories; (2) fit linear and logistic mixed-effects models with random intercepts and slopes; (3) interpret fixed effects, random effects and intraclass correlation; (4) handle missing data and dropout in repeated measures; and (5) know when generalised estimating equations (GEE) are the better choice.

Prerequisites. Level 2 (linear, logistic and count models). Packages: lme4, glmmTMB, geepack, ggeffects.

Session	Content	Product
E1: Repeated measures over time	Long vs wide data; spaghetti plots; why ordinary regression is wrong for repeated measures; linear mixed models for daily measurements (e.g. fluid balance or SOFA on days 1–3); random intercepts and slopes; time as a predictor	Mixed-model analysis of a repeated outcome with a Results paragraph
E2: Clustered outcomes and GEE	Binary repeated or clustered outcomes; logistic mixed models; GEE and population-average vs subject-specific effects; ICC; model checking; reporting	Comparison of mixed-model and GEE estimates for one question

Module F: Complex data: clustered designs, repeated measures and large registries

Aims. Participants will be able to (1) recognise clustering (patients within wards, ICUs or hospitals) and its consequences for standard errors and sample size; (2) analyse cluster-randomised and multicentre data (centre effects, design effect); (3) work with large routine datasets and registries (data linkage, data quality checks at scale, coding systems such as ICD, efficient data handling in R); and (4) plan causal analyses of observational data (DAGs, propensity scores, the target-trial idea, introduced).

Prerequisites. Level 2; Module E recommended. Packages: data.table or arrow, MatchIt, WeightIt, cobalt, dagitty.

Session	Content	Product
F1: Clustered and multicentre data	Design effect and ICC; centre as fixed vs random effect; cluster-robust standard errors; analysing a cluster-randomised or multicentre study; reporting (CONSORT cluster extension)	Analysis plan for a clustered or multicentre dataset
F2: Large registries and routine data	Data linkage and de-identification; data quality at scale; coding systems; efficient data handling; causal questions from routine data (DAGs, propensity scores with matching, weighting and balance checks, and target-trial emulation, introduced); RECORD reporting	Data-management and analysis plan for a registry-based question

Module G: Reproducible research and publication-quality output

This module holds the reproducibility and figure topics moved from Level 2. It is recommended as the first Level 3 module for everyone, because the other modules use its project structure and figure rules.

Aims. Participants will be able to (1) organise an analysis as a reproducible R project that someone else can re-run (the "re-run in a year" test); (2) keep a history of their work with version control; (3) write a Quarto report in which every number, table and figure is produced by code; and (4) produce figures that meet journal requirements.

Prerequisites. Level 2. Packages: here, renv, quarto, gtsummary, flextable, ggplot2, ragg, svglite, patchwork; Git through RStudio or GitHub Desktop.

Session	Content	Product
G1: Reproducible projects and version control	Why reproducibility matters; the pipeline from raw data to clean data, analysis and report; an RStudio Project with a standard folder structure (`data/raw`, `data/clean`, `R/`, `output/`, `report/`); paths with `here`; script headers, numbered scripts, packages loaded at the top, `sessionInfo()` and `renv`; version control with Git (commit, history, restore, a remote copy on GitHub or similar), with dated versions and a change log as the minimum alternative	A FLUID-ICU project under version control that re-runs from the raw file on another computer

Session	Content	Product
G2: Quarto reports and journal-ready figures	A Quarto report that runs the whole pipeline and renders to Word and HTML; inline numbers in sentences; one home for every number; tables in journal style; figures for journals: vector (PDF, EPS, SVG) vs raster (TIFF, PNG) files and when each is required, width and height in mm for single and double columns, resolution (300 to 600 dpi or more for line art, as the journal states), fonts and minimum font sizes, colour-blind-safe palettes, multi-panel figures with `patchwork`, exporting with `ggsave`; checking a figure against a journal's author instructions	A rendered Quarto report and two figures exported to a target journal's requirements

Module H: Missing data and sensitivity analyses in depth

Aims. Participants will be able to (1) build and check a multiple-imputation model (variables to include, the outcome in the imputation model, number of imputations, convergence and imputed distributions, Rubin's rules); (2) impute non-linear terms, interactions and multilevel or longitudinal data with care; (3) plan and run sensitivity analyses for missing-data assumptions (including MNAR scenarios with delta adjustment), outliers, definitions of exposure and outcome, and unmeasured confounding (E-values, introduced); and (4) report missing data and sensitivity analyses briefly and clearly.

Prerequisites. Level 2 (the Session 1 missing-data basics and at least one regression model). Packages: mice, naniar, ggmice, EValue.

Session	Content	Product
H1: Multiple imputation in practice	EDA for missing data (patterns; comparing patients with and without missing values); the imputation model; mice step by step; convergence and imputed-value diagnostics; pooling; complete-case vs imputed estimates (FLUID-ICU Q4: lactate and 28-day mortality)	An imputation analysis with diagnostic plots and a comparison table
H2: Sensitivity analyses	What to vary and why; MNAR scenarios (delta adjustment); outliers and influential points; other definitions of exposure and outcome; unmeasured confounding and the E-value; pre-specifying sensitivity analyses in the SAP; reporting them in a short table	A sensitivity-analysis table with two sentences of interpretation

5. Choosing a module

If you want to…	Module
Summarise all the evidence on a question	**A**: Meta-analysis
Build or judge a risk score, prediction tool or AI model	**B**: Prediction and ML

If you want to…	Module
Test how good a test, score or sign is at detecting a condition	**C**: Diagnostic accuracy
Analyse time-to-event outcomes with competing events or changing exposures	**D**: Advanced survival
Analyse measurements repeated over time on the same patients	**E**: Longitudinal and mixed models
Analyse multicentre, cluster-randomised or registry data	**F**: Complex data
Make an analysis that others can re-run, and figures that meet journal requirements	**G**: Reproducible research (recommended first)
Handle missing data properly and test whether the answer is robust	**H**: Missing data and sensitivity analyses

Key free resources for each module are listed in `Course/References/further_reading.md`: the Cochrane Handbook and PRISMA 2020 (A); TRIPOD+AI and the Riley sample-size paper (B); STARD 2015 and QUADAS-2 (C); *Statistics at Square One* and the Bland–Altman Statistics Notes (D, E); the CONSORT cluster extension (F); the Quarto documentation and the target journal's author instructions for figures (G); the Sterne *BMJ* 2009 paper on multiple imputation (H).